

Operations & Management KPI Dashboard

Track current backlog, on-time completion, service cycle time, rework and delivery cost.

BUSINESS BRIEF | CLIENT REQUIREMENTS | DATA MODEL | STEP-BY-STEP BUILD

What the project does

Track current backlog, on-time completion, service cycle time, rework and delivery cost. It turns the specified source tables into an interactive, filterable view of these business outcomes.

Synthetic demo checkpoints

Reference figures below describe the included synthetic dataset only; they are not client results.

REFERENCE KPI	DEMO VALUE
Jobs Created	720
Completed Jobs	573
Open Jobs	147
On Time Rate	16.1%
Overdue Open Jobs	137

REQUIREMENTS NOTE | This guide translates the repository specification into an implementation brief. Confirm it with the actual client; it is not a signed discovery or acceptance document.

01 / Business problem and client requirements

Agree the outcome and KPI definitions before loading customer data.

Business problem

Track current backlog, on-time completion, service cycle time, rework and delivery cost.

Without consistent definitions, business totals can be difficult to compare or trace back to a responsible team, customer, product, or period. The dashboard is designed to let managers review the specified KPIs, apply business filters, and inspect supporting detail. This is the intended use inferred from the project specification, not a claim about a named client's current process.

Requirements interpreted from the project specification

- Provide a Power BI report that supports the stated goal: Track current backlog, on-time completion, service cycle time, rework and delivery cost.
- Show the 3 project pages, their KPI cards, charts, filters, and detail tables listed in the page blueprint.
- Calculate the named measures using their documented formulas and preserve the project rules.
- Use the reporting scope: Creation-date cohorts and current job status at 30 Sep 2026.
- Allow analysis by: Calendar.Month, Jobs.Service, Jobs.Status, Teams.Location
- Deliver editable PBIP report and semantic-model source, a data contract, private-source templates, and a reconciliation path.

Project boundaries

Month means CreatedDate. CompletedDate is required for Completed and blank for Open. OnTime and Rework are 0/1 and zero for open jobs. CompletedDate >= CreatedDate; CycleDays must match. DueDate is a calendar-date SLA, not business-hour SLA. Update as-of DATE in overdue measure when using a newer snapshot. Historical backlog and completion-date throughput need a separate event history.

Client discovery questions

Confirm source systems and data owners; reporting currency and timezone; period and as-of date; refresh frequency; exact business definitions; privacy and access requirements; who signs off; and whether publishing to Power BI Service is in scope.

02 / Data contract and model

Keep the input tables at their documented grain; the grain determines what a summed KPI means.

TABLE	GRAIN / KEY	FIELDS (SOURCE TYPE)
Teams	one operations team / TeamID	TeamID (string), Team (string), Location (string)
Jobs	one current job snapshot / JobID	JobID (string), TeamID (string), Service (string), Priority (string), CreatedDate (dateTime), DueDate (dateTime), CompletedDate (dateTime), Status (string), CycleDays (int64), OnTime (int64), Rework (int64), Cost (decimal)
Metrics	one metadata row / Label	Label (string)
Calendar	one calendar date / Date	Date (dateTime), Year (int64), Month (string), Quarter (string)

Relationships in the PBIP model

FROM (MANY)	TO (ONE)	FILTERING
Jobs.TeamID	Teams.TeamID	oneDirection
Jobs.CreatedDate	Calendar.Date	oneDirection

Data handoff: Use data/import-template with exact filenames and headers. Keep client source files in approved private storage; do not commit them. Set DataMode=Folder and DataFolder to the approved location only after validating the inputs.

03 / KPI and DAX glossary

Treat these formulas and definitions as a signed calculation contract.

MEASURE	DAX SPECIFICATION	MEANING
Jobs Created	COUNTRAWS(Jobs)	Jobs in selected creation-date cohorts.
Completed Jobs	CALCULATE(COUNTRAWS(Jobs), KEEPFILTERS(Jobs[Status] = "Completed"))	Currently completed jobs in selected creation cohorts.
Open Jobs	CALCULATE(COUNTRAWS(Jobs), KEEPFILTERS(Jobs[Status] = "Open"))	Current unfinished jobs in selected creation cohorts.
Completion Rate	DIVIDE([Completed Jobs], [Jobs Created])	Completed jobs / created jobs in cohort; not daily throughput.
On Time Jobs	SUM(Jobs[OnTime])	Completed jobs by due date.
On Time Rate	DIVIDE([On Time Jobs], [Completed Jobs])	Completed on-time jobs / all completed jobs.
Average Cycle Days	CALCULATE(AVERAGE(Jobs[CycleDays]), KEEPFILTERS(Jobs[Status] = "Completed"))	Mean creation-to-completion days for finished work.
Overdue Open Jobs	CALCULATE([Open Jobs], KEEPFILTERS(Jobs[DueDate] < DATE(2026, 9, 30)))	Open jobs whose due date precedes the demo snapshot date.
Rework Jobs	SUM(Jobs[Rework])	Completed jobs flagged for rework.
Rework Rate	DIVIDE([Rework Jobs], [Completed Jobs])	Reworked completed jobs / completed jobs.
First Pass Yield	DIVIDE([Completed Jobs] - [Rework Jobs], [Completed Jobs])	Completed without rework / all completed jobs.
Recorded Cost	SUM(Jobs[Cost])	Recorded cost incurred on all selected jobs; open jobs included.
Cost per Completed Job	DIVIDE(CALCULATE(SUM(Jobs[Cost]), KEEPFILTERS(Jobs[Status] = "Completed")), [Completed Jobs])	Cost associated with completed jobs / completed count.

Format currency using the client's approved reporting currency. Show rate, margin, attainment, and share measures as percentages. Format units, hours, days, and counts as whole or decimal numbers only where their source supports it.

04 / Build step by step | 1-5

Work through the stages in order and preserve the original demo as your baseline.

1. Confirm the business decisions

Start with: Track current backlog, on-time completion, service cycle time, rework and delivery cost. Confirm who will use the report and what action each page supports. Agree the reporting period, currency, refresh cadence, data owner, KPI definitions, cutoffs, and acceptance owner. The brief here is inferred from this repository spec, not a signed client requirements document.

2. Request the right source extracts

Request the tables at this grain: Teams: one operations team, key TeamID; Jobs: one current job snapshot, key JobID. Use the exact file names, headers, primary keys, and types listed in the data dictionary. Ask for historical periods only where the project scope supports them; do not assume snapshots can reproduce history.

3. Keep data secure and test its quality

Place client files in approved private storage, outside GitHub. Check unique keys, required fields, valid dates, numeric signs, foreign keys, and coverage. Preserve these rules: Month means CreatedDate. CompletedDate is required for Completed and blank for Open. OnTime and Rework are 0/1 and zero for open jobs. CompletedDate >= CreatedDate; CycleDays must match. DueDate is a calendar-date SLA, not business-hour SLA. Update as-of DATE in overdue measure when using a newer snapshot. Historical backlog and completion-date throughput need a separate event history. The demo is synthetic and is for testing the model, not for client reporting.

4. Open the PBIP source project

Download and extract the full repository, then open power-bi/10-operations-kpis/Dashboard.pbip in Power BI Desktop. Keep the complete folder structure. PBIP saves report and semantic-model definitions as project files that can be versioned [1].

5. Check the known-good demo baseline

Refresh in Demo mode. Review DATA-CONTRACT.md, validation/expected-demo.json, and Model.SemanticModel/DAXQueries/Reconcile.dax. Record the supplied demo totals and click through every page before replacing the source.

05 / Build step by step | 6-10

Connect private data, build the model, and test the finished report.

6. Connect the client's approved data

Copy the matching CSV templates from `power-bi/10-operations-kpis/data/import-template/` to the client's private source folder. In Power Query > Manage Parameters set `DataMode` to `Folder` and `DataFolder` to that absolute path. Keep the required Metrics row. Select Transform Data to inspect types and fix errors before loading [2].

7. Prepare fields and relationships

Use the declared data types, retain one row at the required grain, and handle blanks only as documented. Check these many-to-one links in the model: `Jobs.TeamID > Teams.TeamID`; `Jobs.CreatedDate > Calendar.Date`. Lookup tables should filter facts; avoid fact-to-fact joins that multiply values [3].

8. Add and reconcile measures

Use `measures.dax` and the KPI glossary as the calculation contract. Create explicit DAX measures with approved formats. Test results under each date and business-unit filter; verify totals against an independent calculation. Measures respond to the report filter context [4].

9. Build and review the report pages

Recreate the report blueprint below: exact page purpose, slicers, KPI cards, chart series, and detail rows. Add clear titles, sort time labels chronologically, format currency and percentages, and test interactions. Keep project-specific definitions visible to the report user.

10. Sign off, publish, and hand over

Refresh, reconcile source totals, test edge cases, document KPI ownership, and obtain business sign-off. Keep the PBIP source in version control. If service publishing is in scope, publish to the approved workspace, configure credentials and gateway if needed, schedule refresh, and test permissions [5]. Do not publish confidential data before access/security sign-off.

06 / Report blueprint and acceptance

Implement each page to support a specific decision and trace each number to source data.

PAGE / FILTERS	KPI CARDS	CHARTS / DETAIL TABLE
Management scorecard Filters: Month, Location, Service	Jobs Created, Open Jobs, On Time Rate, First Pass Yield	Charts: Completed and open jobs by team; On-time rate by creation month Detail: Team, Jobs Created, Completed Jobs, On Time Rate, Rework Rate
Backlog and service Filters: Month, Location, Status	Open Jobs, Overdue Open Jobs, Average Cycle Days, Completion Rate	Charts: Open and overdue jobs by priority; Overdue open jobs by team Detail: JobID, Team, Service, Priority, Status, CreatedDate, DueDate, Open Jobs, Overdue Open Jobs
Quality and cost Filters: Month, Location, Service	Rework Jobs, Rework Rate, Recorded Cost, Cost per Completed Job	Charts: First Pass Yield by Service; Recorded Cost by Month Detail: Team, Completed Jobs, Average Cycle Days, First Pass Yield, Cost per Completed Job

Acceptance checklist

- Open the PBIP in current Power BI Desktop and refresh all tables without errors.
- Reconcile source counts, key uniqueness, relationships, and every project-specific data rule.
- Compare KPI measures to independently calculated totals and the supplied expected-demo.json.
- Test each slicer, filter interaction, date behavior, chart title, table detail, blank context, and zero denominator.
- Confirm currency, percentage formats, report audiences, privacy, access, refresh ownership, and client sign-off.

Demo acceptance values

METRIC	EXPECTED (SYNTHETIC)
Jobs Created	720
Completed Jobs	573
Open Jobs	147
On Time Rate	16.1%
Overdue Open Jobs	137
First Pass Yield	91.1%
Recorded Cost	PKR 14,209,000

07 / Handover and references

Use the Microsoft guidance below for product workflows; verify tenant-specific settings at delivery.

- Document approved data owner, source, refresh schedule, and support contact.
- Keep the metric dictionary, data contract, sign-off totals, and known limits with the delivered report.
- For Service deployment, confirm target workspace, credentials, gateway needs, permissions, tenant policies, and applicable license.
- A slicer does not enforce row-level security. Design and test security separately before exposing restricted data.
- This repository contains synthetic examples and a source implementation. Desktop rendering, engine evaluation, Service deployment, and client acceptance are separate release gates.

Microsoft Learn references

[1] [Power BI Desktop projects \(PBIP\) - project format and folders](https://learn.microsoft.com/en-us/power-bi/developer/projects/projects-overview)

<https://learn.microsoft.com/en-us/power-bi/developer/projects/projects-overview>

[2] [Connect to data in Power BI Desktop - Get Data and Power Query](https://learn.microsoft.com/en-us/power-bi/connect-data/desktop-connect-to-data)

<https://learn.microsoft.com/en-us/power-bi/connect-data/desktop-connect-to-data>

[3] [Understand star schema and the importance for Power BI](https://learn.microsoft.com/en-us/power-bi/guidance/star-schema)

<https://learn.microsoft.com/en-us/power-bi/guidance/star-schema>

[4] [Tutorial: Create your own measures in Power BI Desktop](https://learn.microsoft.com/en-us/power-bi/transform-model/desktop-tutorial-create-measures)

<https://learn.microsoft.com/en-us/power-bi/transform-model/desktop-tutorial-create-measures>

[5] [Publish semantic models and reports from Power BI Desktop](https://learn.microsoft.com/en-us/power-bi/create-reports/desktop-upload-desktop-files)

<https://learn.microsoft.com/en-us/power-bi/create-reports/desktop-upload-desktop-files>

References checked 30 September 2026. Power BI interface labels, preview features, licensing, and tenant controls can change; confirm them in the client's environment.